Dr. Marc Winstel

CBDC and Financial-Market Infrastructure Researcher

Frankfurt area, Germany
marcwinstel@pm.me
Google Scholar | INSPIRE-HEP

Profile

CBDC and financial-market infrastructure researcher combining central-bank domain expertise, distributed-systems understanding, and technical-policy analysis. Applies a background in theoretical and computational physics to payment systems, CBDC, distributed ledger technology (DLT), settlement infrastructure, and policy-relevant design trade-offs. Experienced in scientific software development, quantitative research, structured argumentation, stakeholder-oriented analysis, and communication of complex technical work through publications, talks, seminars, and teaching.

Professional experience

Researcher *Jul 2025 – present*
Deutsche Bundesbank — Frankfurt am Main, Germany

- Conduct applied research on CBDCs, digital money, distributed systems, and financial-market infrastructure; consensus design trade-offs in payment and settlement systems, offline-functionality evaluation, analysis of DLT and consensus architectures for financial infrastructures.
- Produced three publicly available research outputs within the first year at Deutsche Bundesbank, covering CBDC system design and offline-functionality evaluation, token-based payment algorithms, and conceptual links between CBDC and online social-network structures.
- Provide research-based technical-policy input for internal project steering, translating technical findings into structured analysis for stakeholders.

Graduate research and teaching assistant *Nov 2020 – Jun 2025*
Institute for Theoretical Physics, Goethe University — Frankfurt am Main, Germany

- Designed and conducted analytical and computational research projects on strongly interacting matter under extreme conditions, including self-initiated projects and international collaborations.
- Built numerical tools in C++ and Python for model studies involving high-dimensional minimization, Monte Carlo methods, and data analysis; translated abstract research questions into adaptive analytical and computational workflows.
- Communicated findings through 8 peer-reviewed journal publications, including one single-author paper and one publication in *Physical Review Letters*, as well as presentations at more than 25 conferences and seminars.
- Organized and hosted a joint specialist seminar between DFG research-center subprojects and the local chiral field theory group for 4 semesters, coordinating technical exchange across research groups.
- Contributed to DFG research-center renewal material and review processes in 2021 and 2025; served as referee for *Physical Review D* from 2023 onward.
- Held tutorials and exercise classes across more than 10 semesters for roughly 100 students. Took lead tutor responsibility for 3 lectures, including a mandatory B.Sc. course with over 70 students; coordinated up to 10 tutors, designed exercises, proofread lecture notes, gave substitute lectures, handled student communication, and advised one B.Sc. thesis student.

Education

2020–2025	**Dr. phil. nat., Theoretical Physics**, Goethe University Frankfurt. Graduated with *summa cum laude*; advisor: Prof. Dr. Marc Wagner. Dissertation on spatially modulated regimes in QCD-inspired models at non-vanishing densities.
2018–2020	**M.Sc. Physics, focus on computational physics**, Goethe University Frankfurt. Thesis on inhomogeneous phases in the 2+1-dimensional Gross-Neveu model via mean-field lattice field theory.
2015–2018	**B.Sc. Physics**, Goethe University Frankfurt. Minor: Computer Science.

Selected credentials and output

Awards	Giersch Excellence Award 2025 for an outstanding dissertation; Giersch Excellence Grant 2021 for outstanding scientific work during the master thesis and doctoral project.
Publications	8 peer-reviewed journal publications in theoretical and computational physics, including one single-author paper and one publication in *Physical Review Letters*. Public research outputs at Deutsche Bundesbank on CBDC system design, token-based payment mechanisms, and DLT/consensus governance.